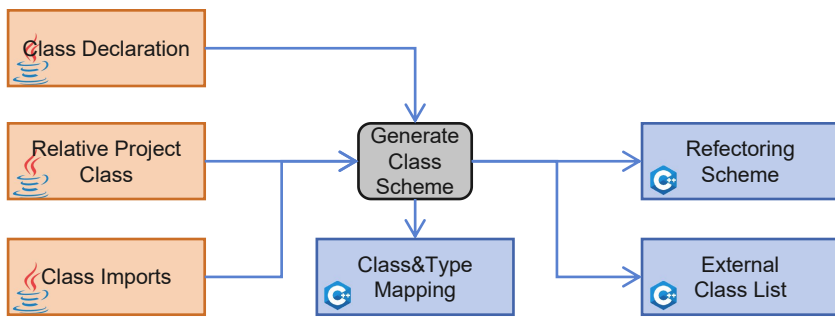




Following is a java file information from the project, now it is supposed to be translated to a C++ header Including declarations of all fields and methods.

{java_class declaration, java imports}

Carefully read the java code, then specify a detailed translation schema, including the following contents:

1. The necessary refactoring to align differences in language features, such as: use a pure virtual base class to replace an interface, use RTTI to replace Java's instanceof operator, add extra fields and methods to imitate reflection, etc.;
2. Any external C++ classes(include enum class, virtual class, etc.) not in the current project and not in standard library need to be implemented in the future;
3. Mapping of data types and classes. The mapped class may be a primitive C++ type, composite type, a class in the C++ Standard Library or a custom class that needs to be implemented;
4. Other additional explanations (if any).

The available headers in the current project are as follows. If you need to use a header that is not included in them, please add it to the list of external classes.

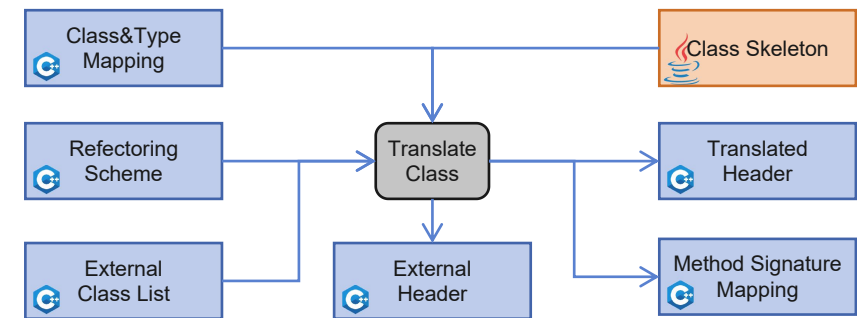
{relative project classes}

The answer should be in the format of json:

{json_example}

Notes:

1. All the above require specific solutions to be provided, DO NOT provide ambiguous answers, and DO NOT directly provide C++ code.
2. For classes in the Java standard library, try to use corresponding classes in the C++ standard library. If there is no corresponding class, add a external class instead.



The following is a part of java file from the source repository. it contains the member variables and member functions(the function implements are omitted). Now it is supposed to be translated to a C++ header file.

{java_skeleton}

The following is the translation scheme:

{scheme}

According to the above translation plan, provide the translated result, your answer should include:

1. The translated code;
 2. Mapping between translated fields/methods and original java fields/methods.
 3. The necessary external class declarations in translation plan(if any), they will be added to the translated repository as additional header files.
- The available header files in the current project are as follows. If you need to use a header file that is not included in them, please add it to the list of external header files.

{header_file_list}

The answer should be in the format of json:

{json_example}

Notes:

1. Only header file is required, don't implement the methods;
2. the "method_mapping" should at least include the following method signatures:

{all_method_signatures}

The following is a part of java file from the source repository. it contains the member variables and member functions(the function implements are omitted). Now it is supposed to be translated to a C++ header file.

{java_skeleton}

The following is the translation scheme:

{scheme}

According to the above translation plan, provide the translated result, your answer should include:

1. The translated code;
 2. Mapping between translated methods and original java methods.
 3. The necessary external class declarations in translation plan(if any), they will be added to the translated repository as additional header files.
- The available header files in the current project are as follows. If you need to use a header file that is not included in them, please add it to the list of external header files.

{header_file_list}

The answer should be in the format of json:

{json_example}

Notes:

1. Only header file is required, don't implement the methods;
2. the "method_mapping" must include the following method signatures:

{all_method_signatures}